

Alejandro Barroso Bueso

alejandrobarrsobueso@gmail.com | +34 684305513 | qrsume.com/barroso

Fourth-year Dual Degree student in Data Science and Telecommunication Engineering with an average grade of 8/10 with hands-on experience in data analysis, embedded systems, and web technologies. Built ML models, conducted security testing, and developed a hardware-based Snake game on STM32. Passionate about solving real-world problems through technology and eager to keep learning in dynamic engineering environments.

EDUCATION

Computer Science and Electrical Computer Engineering Purdue University	2024-2025
Dual Degree in Data Science Engineering and Telecommunication Engineering University Carlos III of Madrid	2022-present
English studies University of Central Florida	2022 (summer)
Dual Diploma (American High School Diploma Online) Academica	2020-2022
Course in JavaScript, PHP, and MySQL Programming University of Salamanca(online)	2020-2022

WORK EXPERIENCE

Creator and designer of web pages, Various Small Businesses Designed and developed versatile websites for small businesses to enhance brand presence and grow client base using data analytics and AI techniques.	2022-present
Math and English teacher, Instituto Torrente Ballester Madrid Worked as an extracurricular English and Mathematics teacher at a high school helping students improve their academic performance and confidence in these subjects.	2023-2024

SKILLS

- Adaptability and Flexibility
- Leadership Skills
- Time Management
- Problem-Solving
- Expert analytical skills
- Teamwork and Collaboration

LANGUAGES

- Spanish (Native)
- English (C1)

PROJECTS

- **Exploiting Web Service Vulnerabilities**
Identification and exploitation of common security flaws in web applications through controlled penetration testing.
- **Embedded Snake Game on STM32 Microcontroller**
Designed and programmed a Snake-style game using buttons, LED matrix, TFT display, and SD card on STM32 implemented logic in C and Assembly for ECE 362 at Purdue University.
- **Exploratory Data Analysis of Titanic Dataset (R)**
Exploratory analysis of Titanic passenger data in R to identify survival patterns based on age, sex, fare, class, and travel conditions.
- **Machine Learning Agent Development: Classification and Regression Analysis**
Designed and implemented machine learning models using Weka to classify optimal snake movements and predict game scores using an automatic gameplay agent

PROGRAMMING LANGUAGES

General Purpose:(Java, Python, C, Bash)

Web Development:(HTML, CSS, JavaScript, PHP, MySQL)

Data Science Analytics:(R, RStudio, SQL, Neo4j/Cypher, Spark, PySpark, Hadoop)

Embedded Systems Hardware:(C, Assembly, Verilog)

